🎮 GAME DEVELOPMENT ROADMAP

📌 Overview

This roadmap guides learners from beginner to advanced levels in game development, covering programming, tools, design, and publishing.

🧩 1. Foundations

Programming

Choose one language:

- C# (Unity)
- C++ (Unreal Engine)
- Python (Basics)

Core Concepts

- Variables, loops, functions
- Object-Oriented Programming (OOP)
- Data structures

Math Basics

- Vectors (movement, direction)
- Coordinate systems (2D/3D)
- Basic physics

🕹️ 2. Game Engine Selection

Choose one engine:

- Unity → Beginner-friendly, C#, 2D & 3D
- Unreal Engine → High-end graphics, C++
- Godot → Lightweight, beginner-friendly

🎯 3. Build Small Games

Start with simple projects:

- Pong
- Snake
- Flappy Bird clone
- Basic platformer

Key Concepts

- Game loop
- Input handling
- Collision detection
- UI basics

🎨 4. Game Design Fundamentals

- Core gameplay mechanics
- Player feedback
- Difficulty balancing
- Level design

🎧 5. Assets (Art & Audio)

Tools

- Aseprite (2D art)
- Blender (3D modeling)
- Audacity (audio editing)

Asset Sources

- itch.io
- OpenGameArt

🧠 6. Intermediate Skills

- Physics systems
- Animation
- AI (NPC behavior, pathfinding)
- Save/load systems
- Basic shaders

Version Control

- Git
- GitHub

🌐 7. Publishing

Platforms:

- Steam
- itch.io
- Google Play Store

🚀 8. Advanced Topics

- Multiplayer networking
- Procedural generation
- Advanced graphics & shaders
- Optimization techniques
- Monetization strategies

👥 9. Portfolio Development

Include:

- 3–5 polished games
- Gameplay videos
- Source code
- Playable builds

⏱ Suggested Timeline

Phase	Duration
Foundations	1–2 months
Small Games	2–4 months
Intermediate	4–6 months
Publishing	6+ months

⚠ Common Mistakes

- Starting large projects too early
- Switching tools frequently
- Not finishing projects
- Ignoring polish

💡 Key Insight

Focus on completing small projects consistently. Iteration builds mastery faster than perfection.